

1. Raw audio waveforms may contain a lot of unnecessary information and are very sensitive to noise, volume, and timing changes. Converting into a spectrogram or MFCC would make important speech patterns and frequency information easier for the model to recognize.
2. It is used during quantization to help TF determine the correct scaling and range of values for converting the model from floating-point numbers to INT8 integers, providing sample input data that is similar to the real data the model will see during inference. It is important because the converter needs to know the typical minimum and maximum activation values to map floating-point values to the smaller INT8 range accurately. Without it, the quantized model may lose accuracy because the scaling factors would not properly represent real input data.
3. It loads and preprocesses audio samples from the speech commands dataset and gets converted into MFCC features before they are used by the model, preparing the input data and labels needed for training, testing, or quantization.
4. `G_model` is the byte array that contains the entire TF Lite model data in hex form, while `g_model_len` stores the size of the model in bytes. The model is converted to this format so it can be embedded directly in the Arduino program memory, since the Nano 33 does not have a standard file system to load .tflite files. TensorFlow Lite Micro uses `g_model` to access the trained keyword spotting model during inference. `G_model_len` specifies the exact size of the model data. If either one is incorrect or incomplete, the Arduino IDE may fail to compile or produce incorrect keyword predictions.
- 5.

```
Heard yes (206) @87232ms
Heard unknown (207) @98560ms
Heard yes (207) @105744ms
Heard unknown (206) @115872ms
Heard no (201) @120944ms
Heard unknown (211) @137136ms
Heard unknown (206) @138896ms
Heard unknown (204) @145056ms
Heard unknown (210) @148192ms
Heard unknown (210) @149760ms
Heard unknown (203) @156944ms
Heard no (207) @161728ms
Heard unknown (203) @163568ms
Heard no (210) @165504ms
Heard unknown (204) @180608ms
Heard unknown (202) @200032ms
Heard unknown (205) @203536ms
Heard unknown (202) @205104ms
Heard unknown (204) @206672ms
Heard yes (212) @211824ms
```